Chapter 4

Primary Memory

Introduction

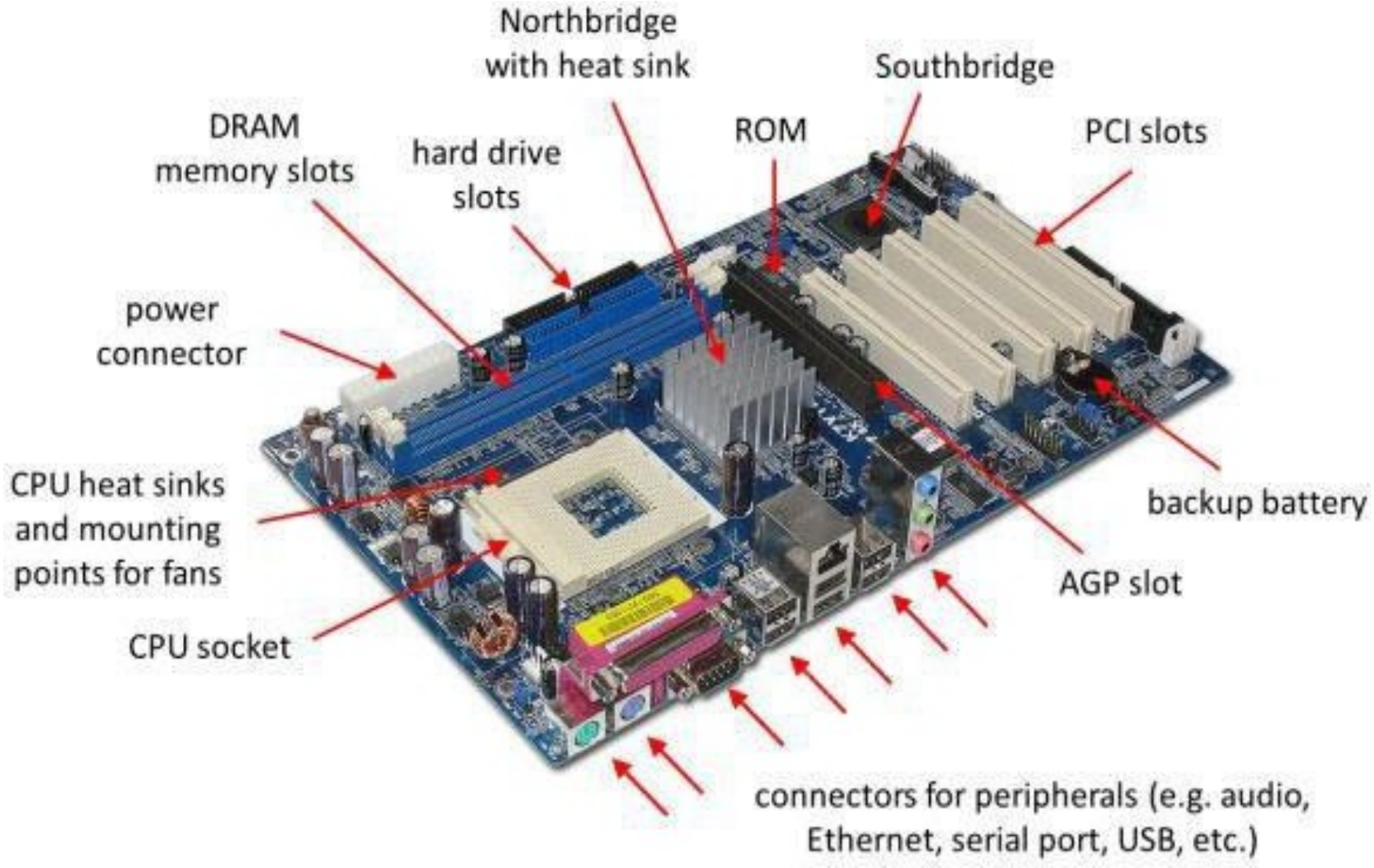
- Computer Memory Refers to electronics holding place for instruction and data where the processor can reach quickly.

- Memory can be classified into two broad categories:

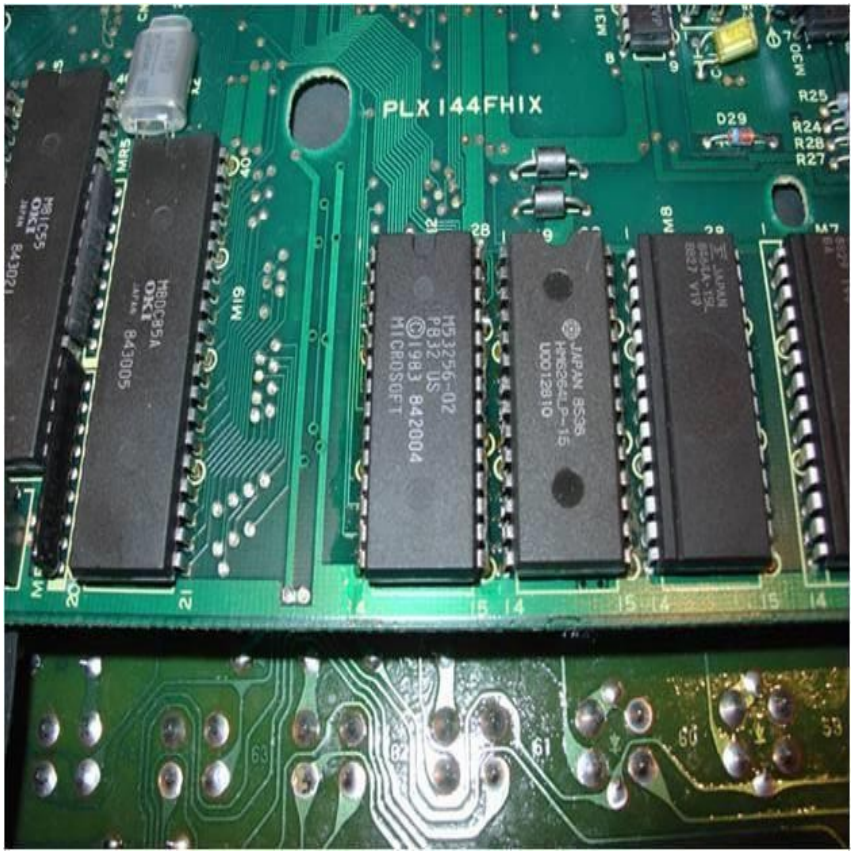
(1) Primary Memory (to handle the data).

(2) Secondary Memory (to store the output).

Motherboard



ROM, RAM


PLX144FHIX
D29
R25
R24
R28
R27
M30
M8
M9
M6
M5
M7
M8C85A
OKI
843005
HM3256-02
P832 US
©1983 842004
MICROSOFT
JAPAN 8538
HM6264LP-15
U001281D


MyService
BL11190.LZ
1GB 200PIN DDR2 SODIMM 128MX64
200935 SU
Removal will void warranty
107069

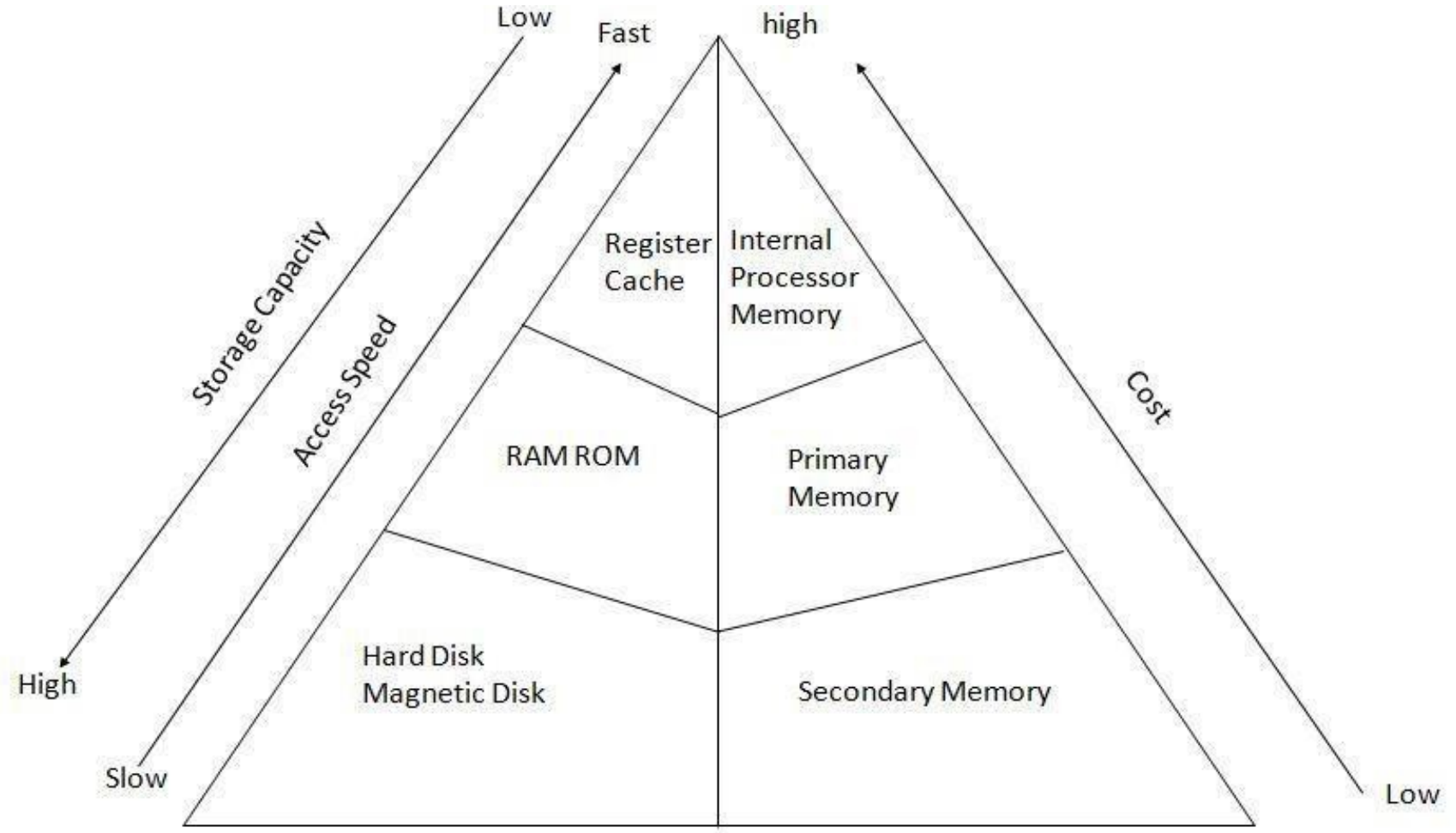
Memory Representation

- The basic unit of memory is bit. Memory can be represented in the bit and bytes..
  - 1 Bit = Binary Digit
  - 8 Bits = 1 Byte
  - 2^10 = 1024 Bytes = 1 KB (**Kilo Byte**)
  - 2^20 = 1024 KB = 1 MB (**Mega Byte**)
  - 2^30 = 1024 MB = 1 GB (**Giga Byte**)
  - 2^40 = 1024 GB = 1 TB(**Terra Byte**)

Memory Hierarchy

- The memory is characterized on the basis of two key factors— capacity and access time.
- Three fundamental types of memory:
  - Internal processor Memory
    - This memory is placed in CPU and it includes cache memory and special registers, both of which can be directly accessed by processor.
  - Primary Memory
    - RAM and ROM fall in the category of primary memory, also known as main memory.
  - Secondary Memory
    - Also known as auxiliary memory, secondary memory provides backup storage for instructions and data.
    - Most commonly used secondary memory devices are hard disk, magnetic disk, and magnetic tapes.

Memory Hierarchy



Storage Evaluation Criteria

- Storage Capacity
  - It refers to size of memory.
- Cost
  - Estimated by the cost per bit of storage.
- Access Time
  - Time required between the request made for read/write operation
    and time it takes for completion of the request.
- Physical Characteristics
  - Four parts namely, electronic, magnetic, mechanical and optical
- Permanence of Storage
- Access Mode
  - Sequential

READ ONLY MEMORY (ROM)

- ROM stands for Read Only Memory.
  - we can only read but cannot write on it.
  - It is non-volatile. The information is stored permanently in such memories during manufacture.
- A ROM, stores such instructions that are required to start a computer. This operation is referred to as bootstrap.
  - ROM chips are in the computer, other electronic items like washing machine and microwave oven.
- BIOS(Basic Input Output System) is the responsible for the startup of computer so it can be considered as a Read only memory.
- ROM for each and Every computer may be what different that's the compatibility issue of platform generating while we use a different platform.
- Also can said OTP(One time programmed) means if it is programmed

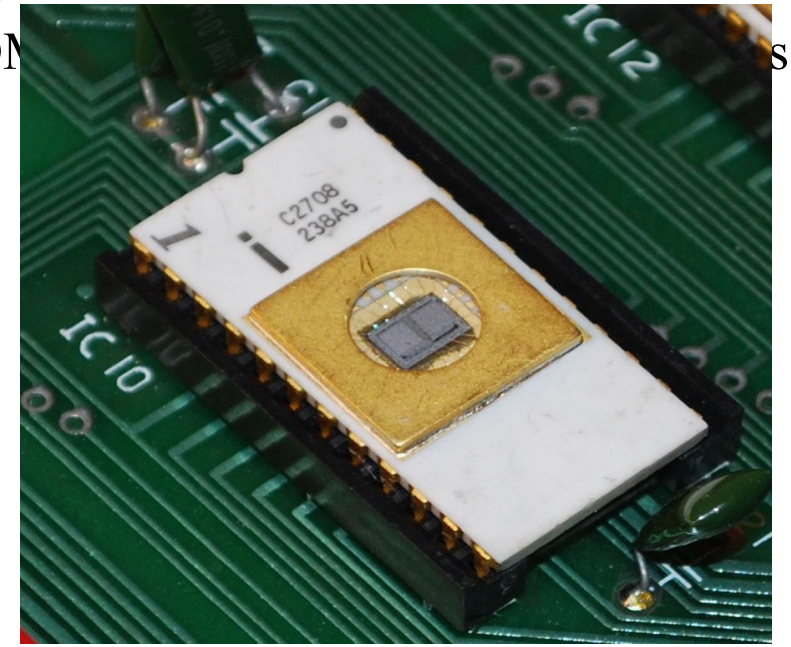
Types of ROM

1. Masked ROM:
   - The very first ROMs were hard-wired devices,
   - It contained a pre-programmed set of data or instructions. These kind of ROMs are known as masked ROMs.
2. Programmable ROM (PROM):
   - The user buys a blank PROM and enters the desired contents using a PROM program, also known as one-time programmable (OTP) device
   - Inside the PROM chip there are small fuses which are burnt during programming.
   - It can be programmed only once and is not erasable.

Types of ROM (cont..)

3. Erasable Programmable ROM (EPROM):

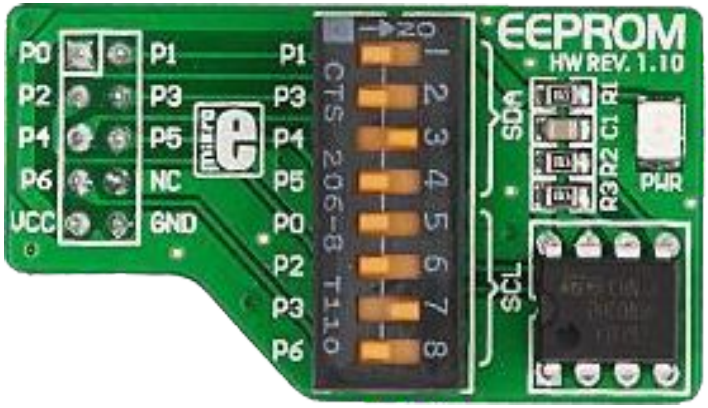
- EPROM can be erased by exposing it to ultra- violet light for a duration of up to 40 minutes.
- This process will erase the entire EPROM and then it reprogrammed.
- EPRO



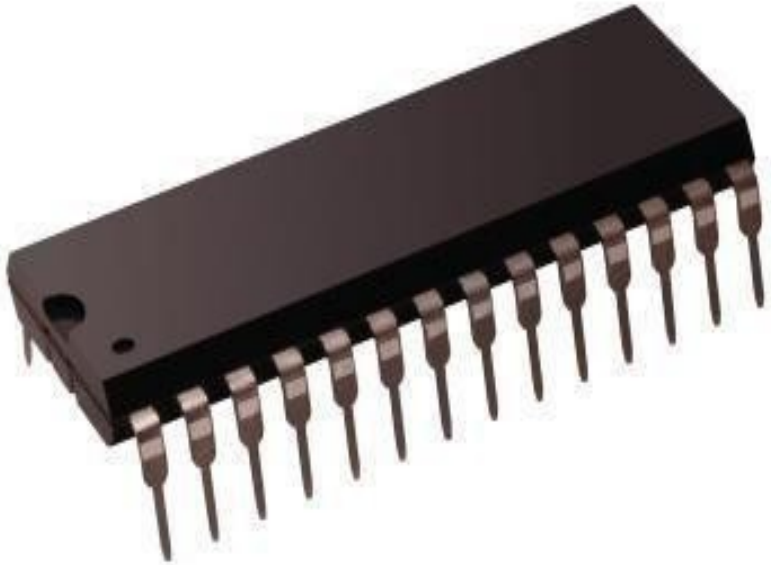
s.

Types of ROM (cont..)

4. Electrically Erasable and Programmable Read Only Memory (EEPROM):
   - EEPROM can be erased by an electrical charge and then written to by slightly higher-than-normal voltage.
   - In EEPROM, any location can be selectively erased and programmed.
   - EEPROMs can be erased one byte at a time, rather than erasing the entire chip. Hence, the process of re-programming is flexible but slow.
5. Flash ROM:
   - Flash ROM, also called flash BIOS or flash memory, is a type of constantly powered non- volatile memory that can be erased and re-programmed in blocks.
   - It is a variation of EEPROM.


EEPROM
HW REV. 1.10
P0
P1
P2
P3
P4
P5
P6
NC
VCC
GND
SDA
SCL
PWR
R3 R2 C1 R1
CTS 206-8 T110


FLASH
MEMORY
CARD
XSO-SCAN30



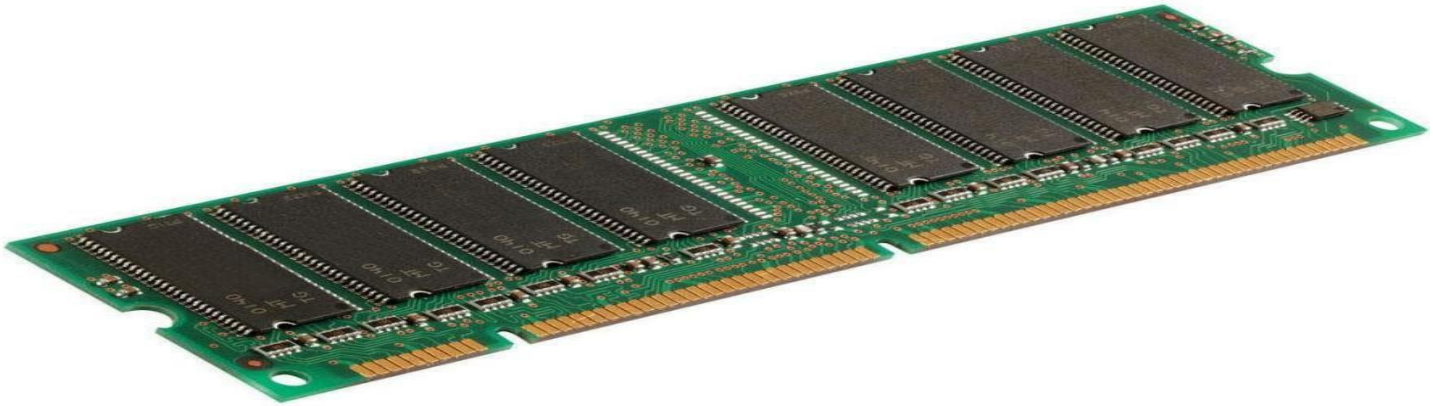

AMIC

Advantages of ROM

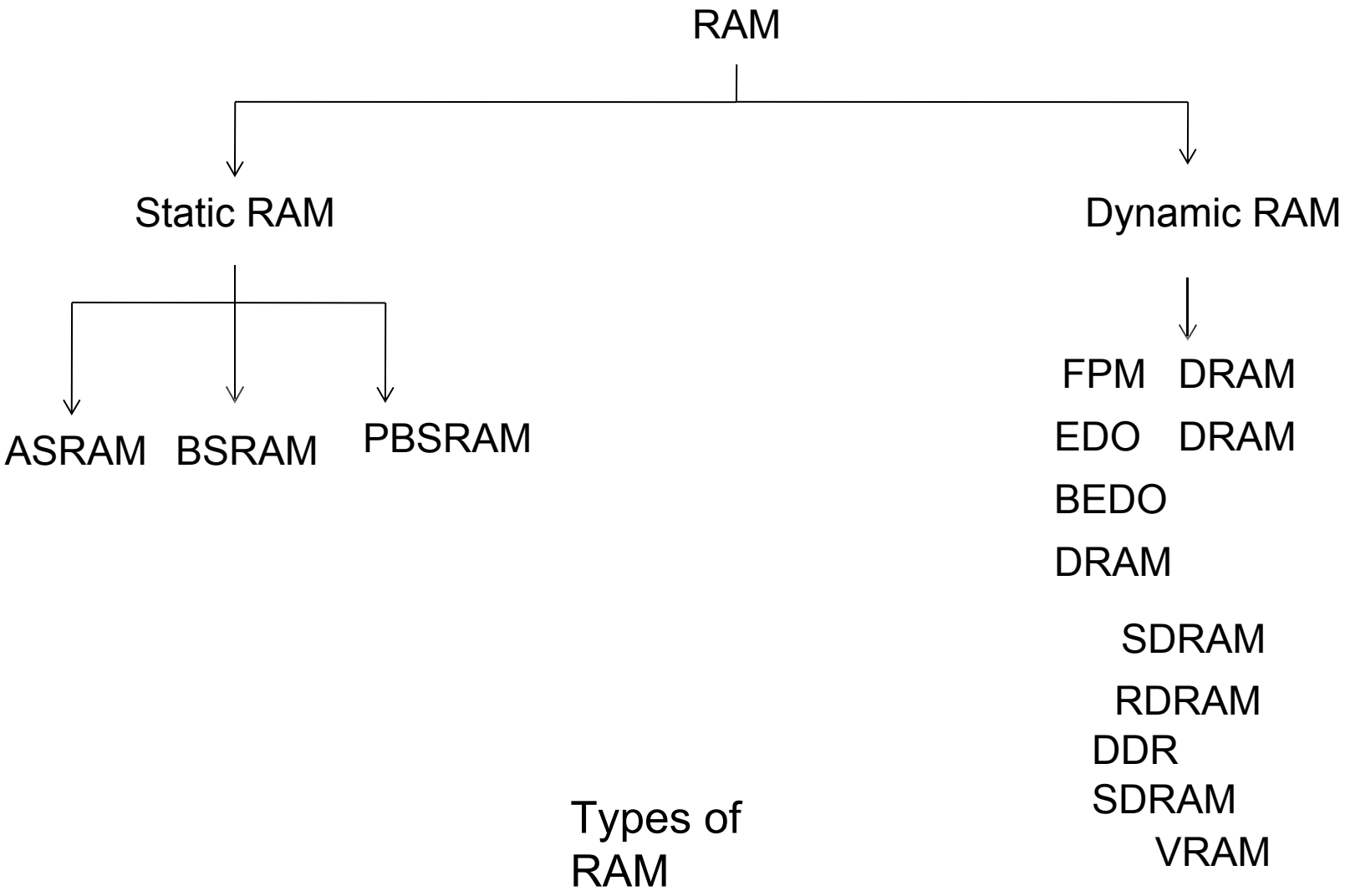
- The advantages of ROM are as follows:
  - Non-volatile in nature
  - These cannot be accidentally changed
  - Cheaper than RAMs
  - Easy to test
  - More reliable than RAMs
  - These are static and do not require refreshing
  - Its contents are always known and can be verified

Random Access Memory (RAM)

- RAM(Random Access Memory) is the primary memory of the CPU for
  - storing data, program and program result.
    - It is read/write memory which stores data until the machine is working. As soon as the machine is switched off, data is erased.
- RAM is volatile, i.e. data stored in it is lost when we switch off the
  - computer or if there is a power failure.
- RAM is small, both in terms of its physical size and in the amount of data
  it can hold
- Data in



Types Of RAM


RAM
Static RAM
Dynamic RAM
ASRAM
BSRAM
PBSRAM
FPM DRAM
EDO DRAM
BEDO
DRAM
SDRAM
RDRAM
DDR
SDRAM
VRAM
Types of
RAM

Static RAM

- The word **static** indicates that the memory retains its contents as long as power is being supplied.
- However, data is lost when the power gets down due to volatile nature.
- It does not need to be **refreshed** periodically.
- SRAM chips use a matrix of 6-transistors and no capacitors.
- SRAM is very fast but more expensive than DRAM.
- SRAM uses more chips than DRAM for the same amount of storage space, thus making the manufacturing costs higher.
- It is often used as cache memory due to its high speed.

DRAM

- DRAM, unlike SRAM, must be continually **refreshed** in order to maintain the data.
- This is done by placing the memory on a refresh circuit that rewrites the data several hundred times per second.
- DRAM is used for most system memory because it is cheap and small.
- All DRAMs are made up of memorycells which are composed of one capacitor and one transistor.

SRAM versus DRAM

- Primary difference => lifetime of the data they store.
  - SRAM retains its contents as long as power is applied on the chip.
  - DRAM has an extremely short data lifetime.
- Total number of cells in the SRAM chip is less than DRAM
- SRAM is fast, has low latency (time gap between request and the action being performed), and does not need to be refreshed.
- SRAM is large and expensive. It require more power to operate so it produces a lot of heat.

SRAM versus DRAM

- DRAM is simple, small and space efficient. It is slower and have longer latency than SRAM.
- SRAM has lower access time as compared to DRAM.

Main Memory

Main Memory is often known as **Primary memory.** It is not like human memory because data stored in this memory, automatically flushes out with power cut off and can not be recovered later. So it is also known as **volatile** memory. It is the workplace of a computer system where data is to be kept on temporary basis for the purpose of processing. Every memory location has a definite address. The total time taken to access data from memory is called **Memory *Access Time* .**

Characteristics of Main Memory

- These are semiconductor memories.
- Usually volatile memory.
- Data is lost in case power is switched off.
- It is the working memory of the computer.
- Faster than secondary memories.
- A computer cannot run without the primary memory.

Secondary Memory

Secondary Memory is also known as external memory or non-volatile memory. It is slower than the main memory. These are used for storing data/information permanently. CPU directly does not access these memories, instead they are accessed via input-output routines. The contents of secondary memories are first transferred to the main memory, and then the CPU can access it. For example, disk, CD-ROM, DVD, etc.

Characteristics of Secondary Memory

- These are magnetic and optical memories.
- It is known as the backup memory.
- It is a non-volatile memory.
- Data is permanently stored even if power is switched off.
- It is used for storage of data in a computer.
- Computer may run without the secondary memory.
- Slower than primary memories.

Difference between RAM and ROM

RAM	ROM
1. Temporary Storage.	1. Permanent storage.
2. Store data in MBs.	2. Store data in GBs.
3. Volatile.	3. Non-volatile.
4.Used in normal operations.	4. Used for startup process of computer.
5. Writing data is faster.	5. Writing data is slower.

Difference between RAM and ROM

Difference between RAM and ROM

Note: The most important things to understand about RAM are that RAM memory is very fast, it can be written to as well as read, it is volatile (so all data stored in RAM memory is lost when it loses power) and, finally, it is very expensive compared to all types of secondary memory . Read-only memory (**ROM**) is a type of non-volatile memory used in computers and other electronic devices. Data stored in **ROM** cannot be electronically modified after the manufacture of the memory device. **RAM** is **costlier** than **ROM** as it involves more complex technology to store information & also it is very much faster than **ROM**.

Storage Unit

A **secondary storage** device refers to any non-volatile **storage** device that is internal or external to the **computer**. It can be any **storage** device beyond the primary **storage** that enables permanent data **storage**. A **secondary storage** device is also known as an auxiliary **storage** device or external **storage**It is required to store data in computer for the purpose of processing in future.Because Primary memory has its own limitations, secondary storage or storage unit is very much required.Secondary storage is used to store large amount of data.Various secondary storage devices are available in market now a days -

–Hard disk (Internal/External)

–Pen drive

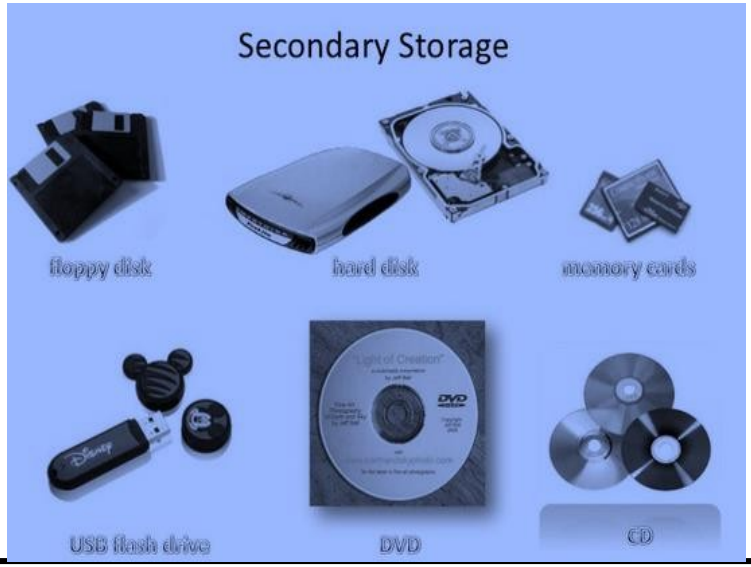
–CD

–DVD

–Blue Ray Disks

–Memory Cards

–Magnetic Tapes

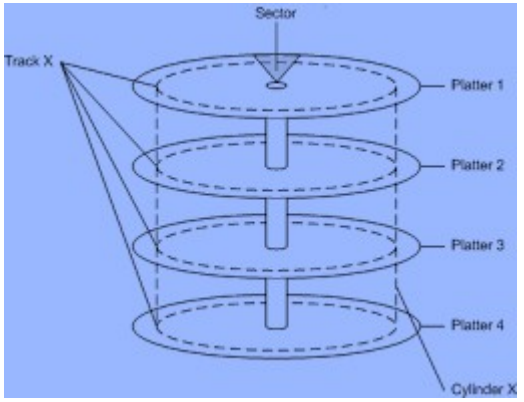


Hard Disk

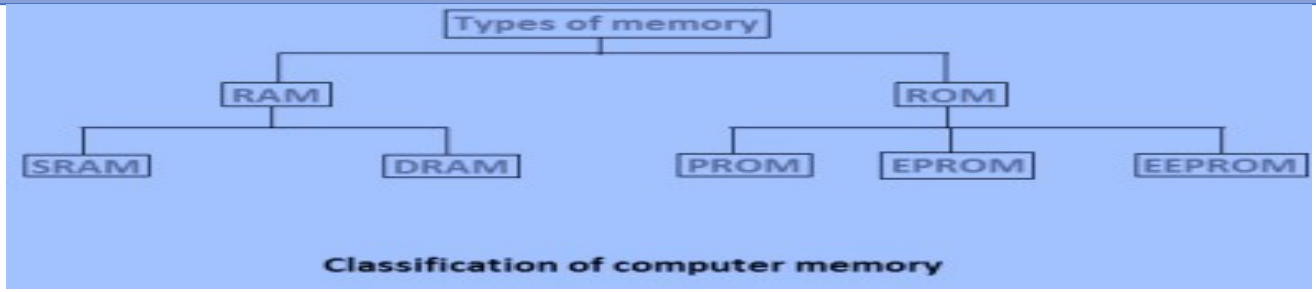
- **Hard Disk Drives** are commonly used as the main storage device in a computer. HDDs often store operating system, software programs and other files. It is a **non-volatile memory** hardware device that controls the positioning, reading and writing of the hard disk.
- A hard disk is a sealed unit containing a number of platters in a stack. It can be mounted in a horizontal or a vertical position and electromagnetic read/write heads are positioned above and below each platter. As the platters spin, the drive heads move in toward the center surface and out toward the edge. In this way, the drive heads can reach the entire surface of each platter.
- Each disk consists of platters, rings on each side of each platter called **tracks**, and sections within each track called **sectors**. A sector is the smallest physical storage unit on a disk, almost always 512 bytes in size.







Memory classification



Classification of computer memory

Main Memory or primary memory is divided into two types-

–RAM (Random Access Memory)

–ROM (Read Only Memory)

The memory which is in direct contact of CPU is known as primary memory. RAM and ROM both remains in direct contact of CPU.

Random Access Memory (RAM) –

It is also called as *read write memory* or the *main memory* or the *primary memory*. The programs and data that the CPU requires during execution of a program are stored in this memory.It is a volatile memory as the data loses when the power is turned off. Relatively cheap per megabyte stored compared to ROM, but relatively expensive compared to secondary memory. It is not possible to imagine processing without RAM.

The efficiency of RAM depends upon the size (capacity)of RAM.RAM supports both read and write operations.

RAM is further classified into two types- **SRAM (Static Random Access Memory) and DRAM (Dynamic Random Access Memory).**

DRAM AND SRAM

- **DRAM** basically consists of transistor and capacitor which stores electric charge.As per switching of Transistor, capacitor may contain a charge (1 bit) or may not contain a charge(0 bit). DRAM needs to be refreshed again and again which is a tedious job for processor.
- **SRAM** is made up of internal Flip-Flops where 1 Flip-Flop is capable of storing 1 bit.It occupies more space as compared to DRAM.It is fast as compared to DRAM.SRAM is expensive as compared to DRAM.

Volatile memory is computer memory that requires power to maintain the stored information. Most modern semiconductor volatile memory is either Static RAM or dynamic RAM.

Static RAM retains its contents as long as the power is connected and is easy to interface to but uses six transistors per bit. **Dynamic RAM** is more complicated to interface to and control and needs regular refresh cycles to prevent its contents being lost. However, DRAM uses only one transistor and a capacitor per bit. SRAM is used for cache memories , where DRAM is used as main memory.

INPUT DEVICES

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Input Devices

- Device: is an instrument that performs a simple task.
- Input: something put into a system.

- An **input device** is any peripheral used to provide data and control signals to a computer.

- An input device is any tool device for entering information into a computer.

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List of Input Devices

- Keyboard
- Mouse
- Trackball
- Joystick
- Light Pen
- Digitizer

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List of Input Devices

- Scanner
- Optical Character Recognition (OCR)
- Optical Mark Reader
- Optical Bar Reader
- MICR (Magnetic Ink Character Recognition)

Keyboard

FU NCTION KEY

NUMERI C PAD



MODIFIE R KEY

AIPHANUMERI C KEY

CURSOR MOVEMEN T KEY

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How Keyboard Works

- Keyboard is a input device used for entering data into computer by pressing one key at one time.

- It has its own processor and circuitry that carries information to and from that processor.

- The key matrix is a grid of circuits underneath the keys.



Microprocessor & Controller Circuitry of Keyboard

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- When you press a key, it presses a **switch**, completing the circuit and allowing a tiny amount of current to flow through.

- It tells the processor the position of each key in the matrix and what each keystroke or combination of keystrokes represents.



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How Keyboard Works

- Most keyboards have between 80 and 110 keys, including: Typing keys ,a numeric keypad, function keys and control keys.
- Using a keyboard, a person can type a document, use keystroke shortcuts, access menus, play games and perform a variety of other tasks.
- Keyboards can have different keys depending on the manufacturer, the operating system they're designed for, and whether they are attached to a desktop computer or part of a laptop.
- But for the most part, these keys, also called **keycaps**,

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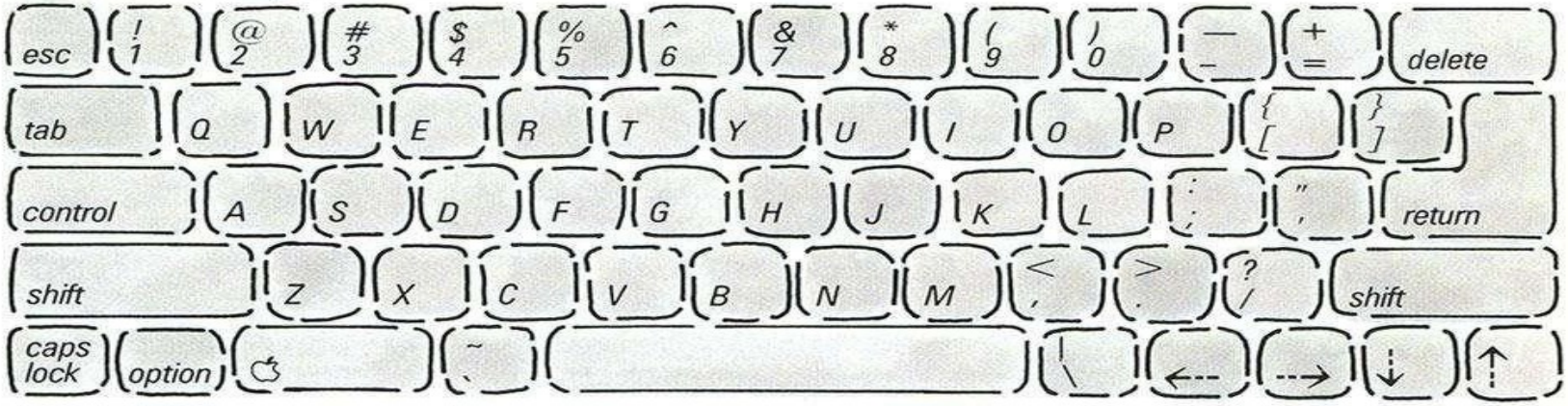
Types of Keyboard (Based on Layout)

- QWERTY Layout
- DVORAK Layout
- ABCDE
- XPeRT
- AZERTY

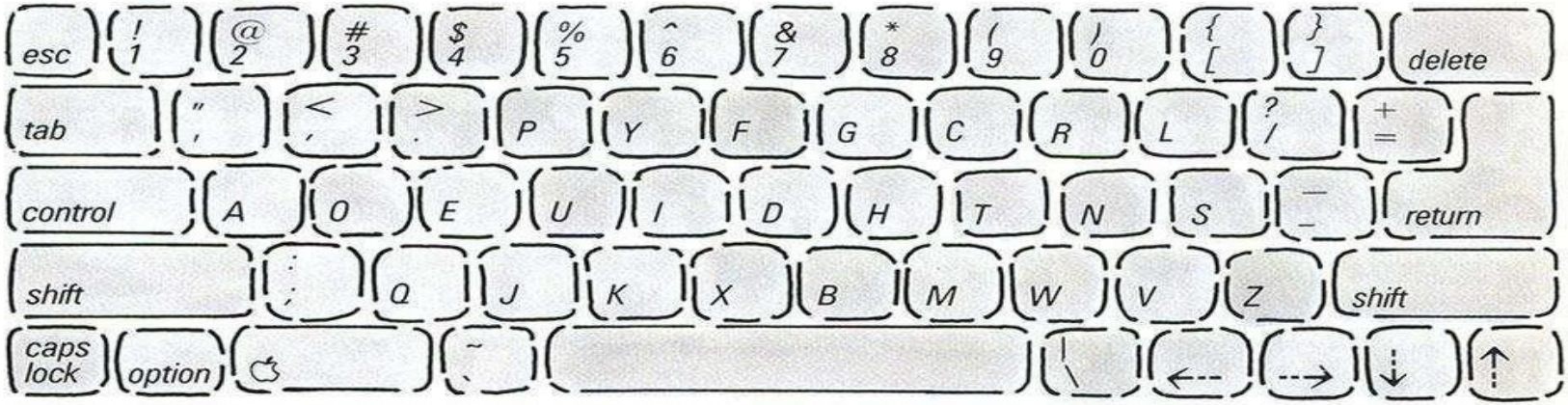
Each is named for the first keys in the pattern. The QWERTZ and AZERTY arrangements are commonly used in Europe.

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Standard Keyboard



Dvorak Keyboard



Source: Apple IIc Plus Owner's Guide, Copyright 1988, Apple Inc.

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Advantages of Keyboard

- Reliable for data input of text and numbers.
- Usually supplied with a computer so no additional cost.

Disadvantages of Keyboard

- Users may be slow for not very accurate typists.
- Slow for accessing menus etc. and difficult to use if you want to move objects around the screen.
- Difficult for people unable to use keyboards through paralysis or muscular disorder.

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MOUSE

- The **mouse** is used to control the movement of a pointer on the screen when it is moved horizontally over a flat surface.

- A ball under the mouse rotates when it is moved and turns two rods, one for left/right and one for up/down.

- Buttons on the mouse let you select options from menus and drag objects around the screen etc. Some models are now wireless.

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Types of Mouse

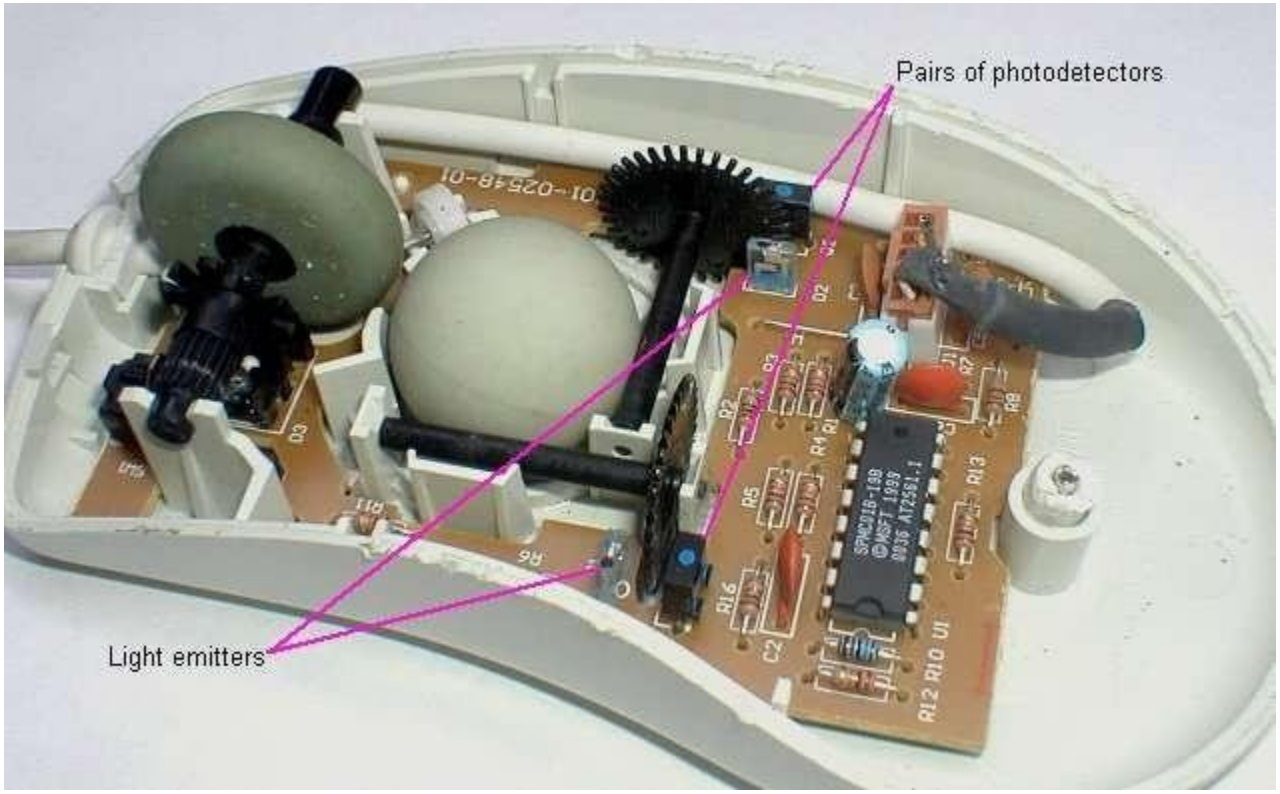
- Mechanical Mouse
  - A mouse that uses a rubber ball that makes contact with wheels inside the unit when it is rolled on a pad or desktop.
- Optical Mouse
  - A mouse that uses light to detect movement. It emits a light and senses its reflection as it is moved,

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OPTICAL MOUSE



MOUSE



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Advantages

- Moves cursor around the screen faster than using keystrokes
- Usually supplied with a computer so no additional cost.
- All computer users tend to be familiar with using them.

Disadvantages

- They need a flat space close to the computer.
- Requires moving hand from keyboard to mouse and back.

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TRACK BALL

- A **trackball** is a pointing device consisting of a ball held by a socket containing sensors to detect a rotation of the ball about two axes— like an upside-down mouse with an exposed protruding ball.

- Instead of moving the whole mouse around, the user rolls the trackball only, which is on the top or side.

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TRACK BALL





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Advantages of Trackball

- Ideal for use where flat space close to the computer is limited.
- Can be useful with laptops as they can be built into the computer keyboard or clipped on.

Disadvantages of Trackball

- Not supplied as standard so an additional cost and users have to learn how to use them.

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JOYSTICK

- A **joystick** is an input device consisting of a stick that pivots on a base and reports its angle or direction to the device it is controlling.

- Joysticks are often used to control video games, and usually have one or more push-buttons whose state can also be read by the computer.

- Joysticks are often used for playing computer games such as flight simulators. They can also be used to control the movement of a wheelchair or other machinery.

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JOYSTICK



- **Advantages of Joystick**
- Easy to learn to use. Very simple design so they can be inexpensive.

- **Disadvantages of Joystick**
- Control can be a bit crude as the directions in simple joysticks are limited to forward, backwards, left and right.
- Better models offer diagonal movement or better.

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Light Pen

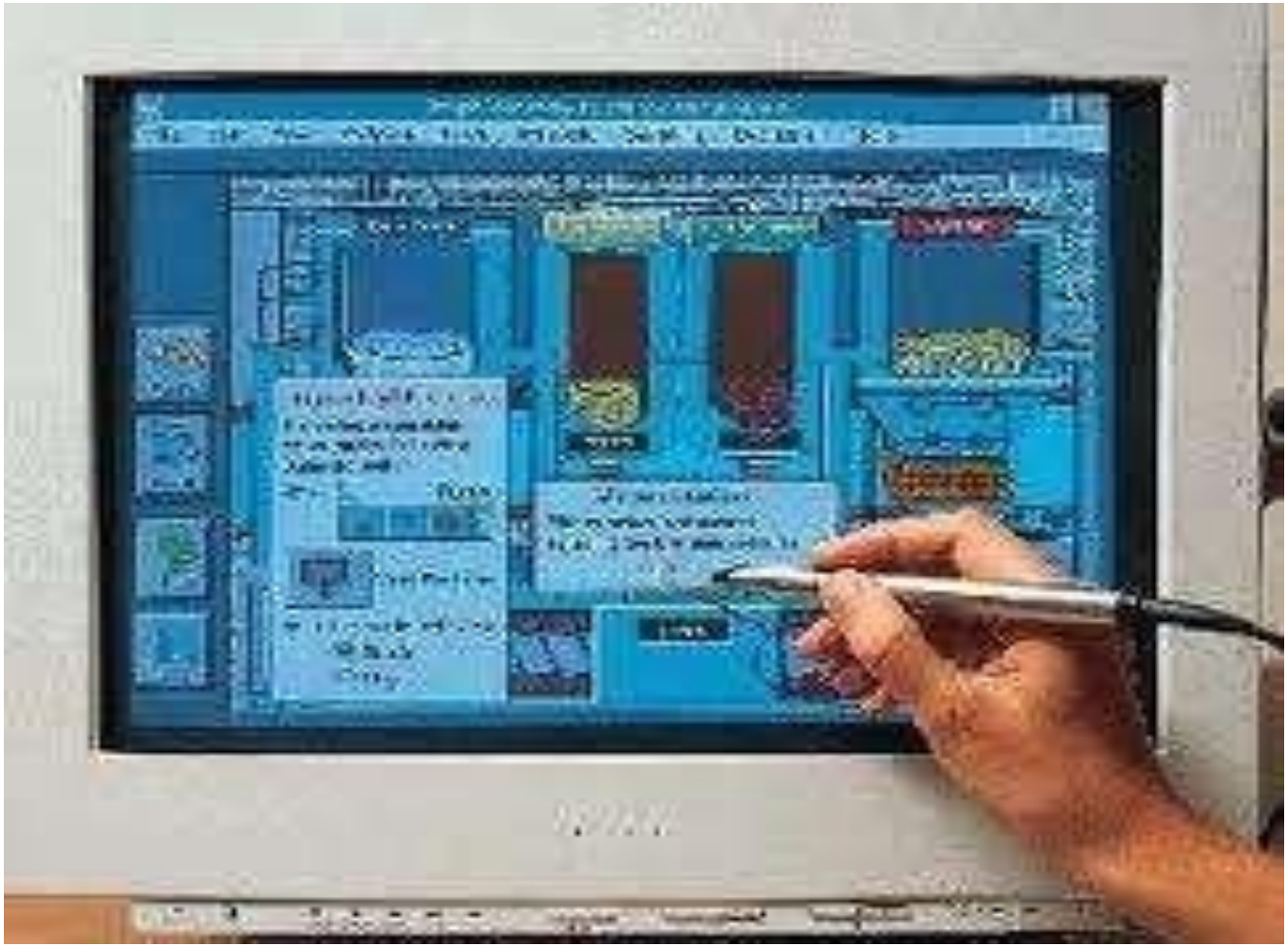
- A **light pen** is a computer input device in the form of a light-sensitive wand used in conjunction with a computer's CRT TV set or monitor.

- It allows the user to point to displayed objects, or draw on the screen, in a similar way to a touch screen but with greater positional accuracy.

- A light pen can work with any CRT-based display, but not with LCD screens ,projectors and other display devices.

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LIGHT PEN



DIGITIZER

- A digitizer is a input device used for converting pictures, maps and drawing into digital form for storage in computer.





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Touch Screen

- A **Touch screen** is an electronic visual display that can detect the presence and location of a touch within the display area.
- The term generally refers to touching the display of the device with a finger or hand.
- Touch Screens can also sense other passive objects, such as a stylus. However, if the object sensed is active, as with a light pen, the term touch screen is generally not applicable.

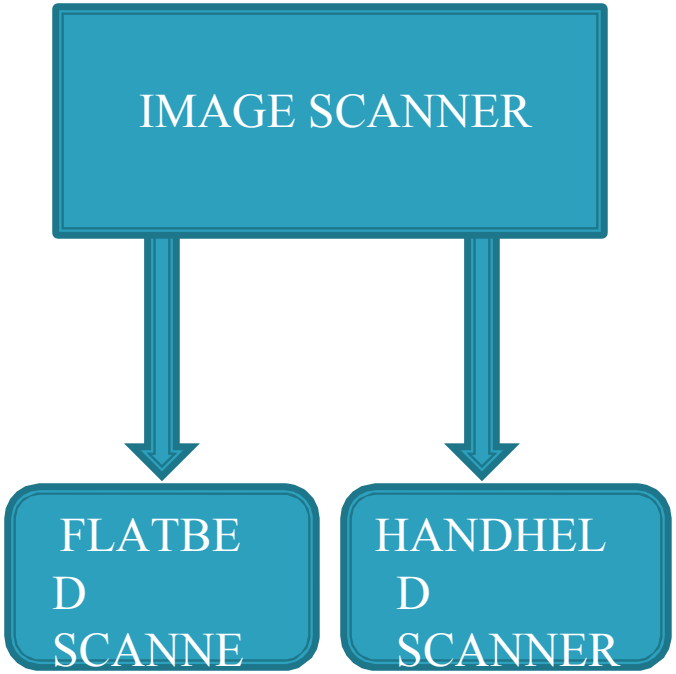
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Touch Screen



Scanning Devices

- Scanning devices are input devices used for direct data entry in to a computer system.



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Flatbed Scanner

- A flatbed scanner is a copier machine consist of a box having a glass plate on the top and lid that cover the glass plate.



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Handheld Scanner

- A hand held scanner has a set of light emitting diode encased in small case that can be held in hand.

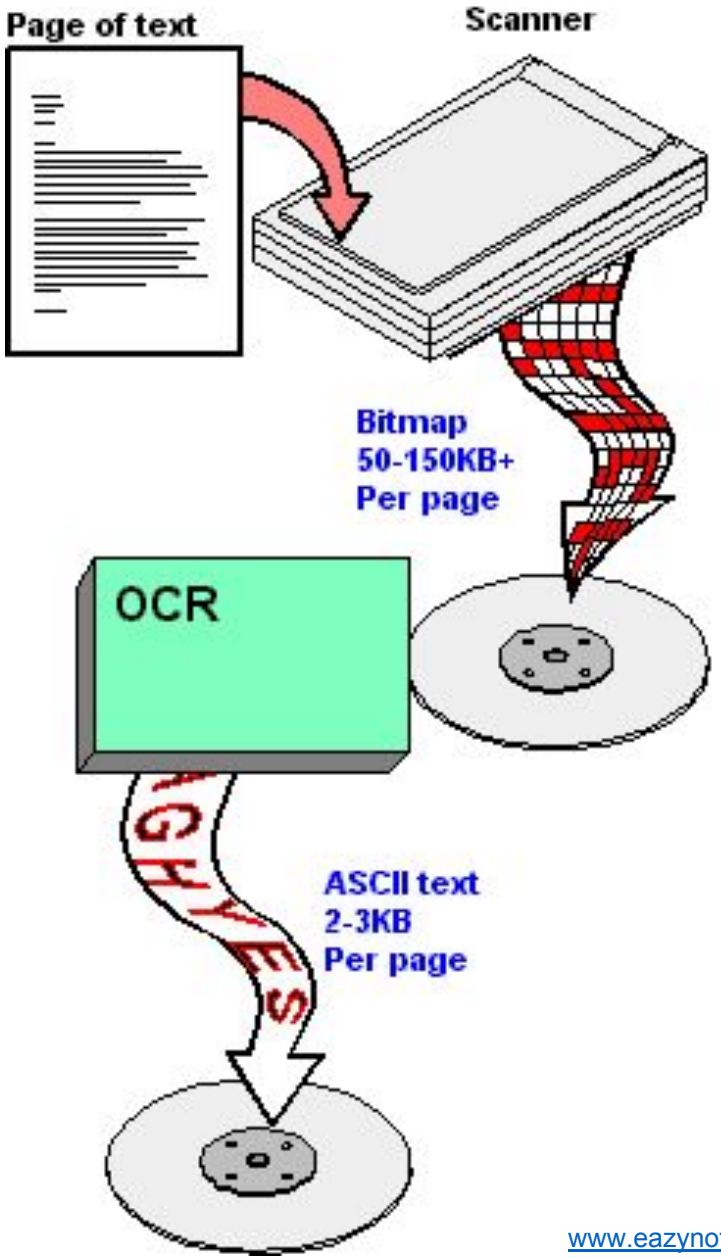


Optical Character Recognition

- OCR is used to convert the bitmap images of a character into ASCII codes
- It can recognize many different OCR fonts, as well as typewriter & computer-printed characters.
- Advanced OCR systems can recognize hand printing.
- OCR software analyzes the light and dark areas of the bitmap in order to identify each alphabetic letter and numeric digit.
- Hand printing is much more difficult to analyze than machine-printed characters.

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From Computer Desktop Encyclopedia
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OCR-A FONT





OCR-B FONT

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Optical Mark Reader

- OMR are used to recognizing a pre-specified type of mark made by pencil or pen.



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Please write your name in the boxes, as you would like it to appear on your Climber certificate and then fill in the corresponding circles below.
*Leave a blank line between first and surname.

Example

| | | |

Bar Code Reader

- **Bar code readers can read bar codes—patterns of printed bars.**

- Lines of different widths and sizes representing data that, when read, will determine what the scanned object is. Bar codes are often used to help organize and be able to index information or prices about an object.
- Barcode readers and scanners

- A **barcode reader** or **scanner** also known as a **point of sale (POS) scanner** are hardware devices capable of reading a barcode and printing out the details of the product and/or logging that product into a database so a company can easily keep track of its inventory
- .

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Bar Code Reader

- A barcode reader consists of a scanner , a decoder , & cable used to connect the reader with a computer.
- Data coded in the form of small line are known as bar code.
- It directs a beam of light across the bar code & measures the amount of light that is reflected back.
- The scanner converts the light energy into electrical energy, which is then converted into data by decoder & forwarded to a computer.

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Magnetic Ink Character Recognition

- MICR is used in banking industry for faster processing of a large number of cheques daily.
- MICR characters are printed in special typefaces with a magnetic ink or toner.

MICR E13 B Font



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Magnetic Ink Character Recognition

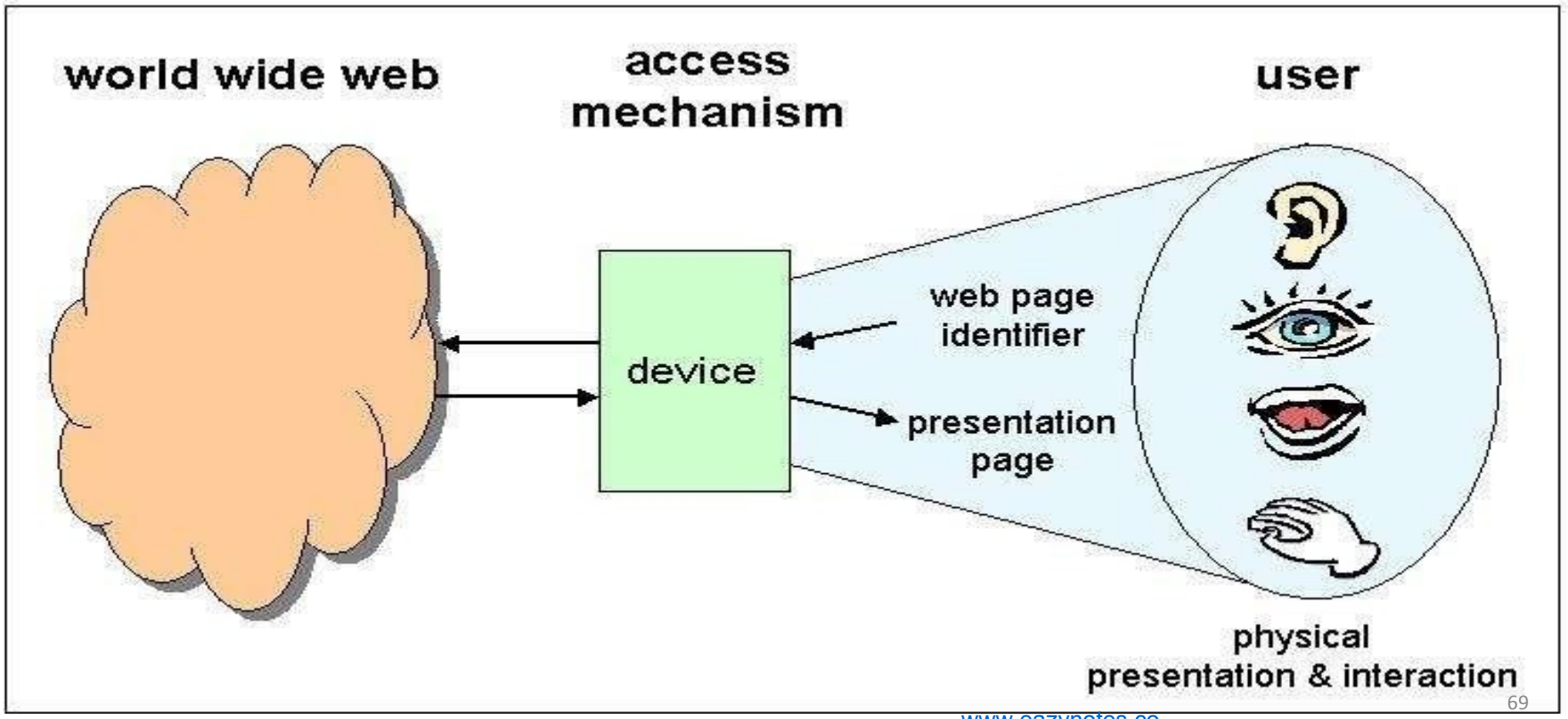




Barcode Printing

Voice Recognition Devices

- It allows a person to input data to a computer system by speaking to it.



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Output Devices

List of Output Devices

- Visual Display Unit
- Printers
- Plotters
- Speakers

Output Devices

- Devices that display, print, or transmit the results of processing from the computer's memory.
- Before an output can be produced by a computer it must have an output device connected to it.

Visual Display Unit (VDU)

- A visual Display Unit (VDU) or Monitors are the most popular output devices used for producing soft-copy output.

- Monitors can display graphics, text and video. The size of a monitor is measured in inches diagonally across the screen; 15, 17, 19 and 21 inch monitors are the most common sizes. A color monitor can display from 16 to over 1 million different colors.



Visual Display Unit (VDU)

Printers

- A printer is an output device that produces a printout or hard copy of the output from a computer. There are many types of printer available.

- The speed of printers can be measured in characters per second (CPS) or pages per minute (PPM), the higher the values are, the faster the printer can produce output.

TYPES OF PRINTERS

- **Impact Printers:** An impact printer forms characters and graphics on a piece of paper by striking a mechanism against an ink ribbon that physically contacts the paper.

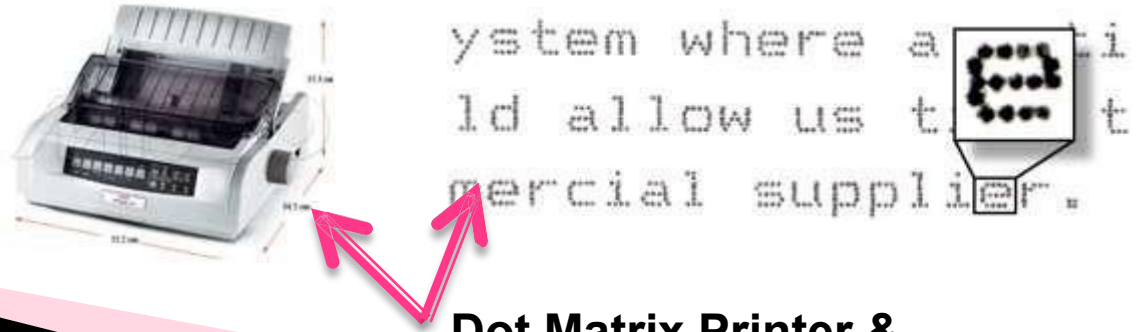
- **Non-Impact Printers:** A non-impact printer forms characters and graphics on a piece of paper without actually striking the paper.

TYPES OF PRINTERS

- Daisy Wheel Printer
- Dot-Matrix Printer
- Line Printer
- Ink-Jet Printer
- Laser Printer
- Plotter

Dot Matrix Printer

- A dot matrix printer forms characters and graphics on the paper by producing pattern of dots.
- Dot matrix printers are relatively cheap and so is the stationery that they use.
- Dot-matrix printers are generally noisy.
- Slow, with speeds usually ranging between 30 to 600 characters per second.



Dot Matrix Printer & Printout

Inkjet Printer

- This is a non-impact printer that forms characters and graphics by spraying tiny drops of liquid ink onto a piece of paper.
- Inkjet Printers are very quite to operate and can produce good quality printouts of both graphics and text.
- Inkjet printers are generally quiet.
- Inkjet printers can produce high quality color output.



Inkjet Printer

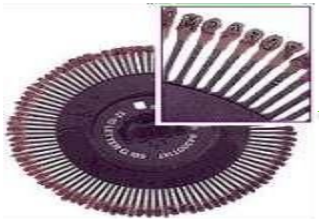
Daisy Wheel Printer

- A daisywheel printer is an impact printer that uses a wheel as a print head. As the wheel rotates, a hammer strikes the backside of the spoke and presses it against the paper to print a character.

- Can print letter quality characters.

- Printing speed is very slow.





Daisy Wheel Printer

Laser Printer

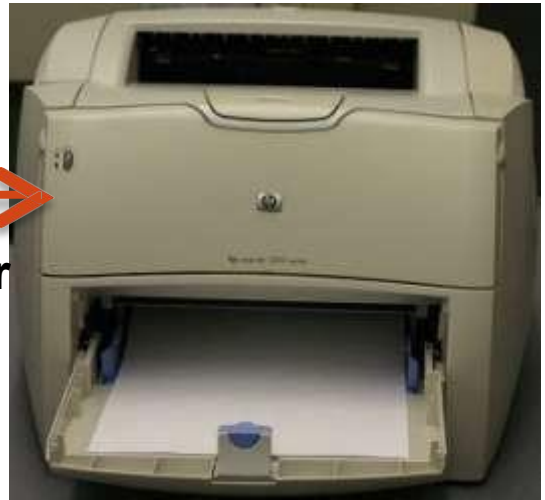
- The mechanism of how a laser printer works is similar to a photocopier. Laser printers are also known as *page printers* because they process and store the entire page before they actually print it.

- Laser printers can print text at speeds of four to over thirty pages per minute.

- The cost per page of toner cartridges is lower than other printers.

- Laser printers are quiet and fast.

Laser Printer



Plotter

- A plotter is a large printer that generates high-quality documents by moving ink pens over
  the surface of a page.

- Plotters areparticularly useful to engineers and architects, as
  they produce high-quality blueprints, maps, and floor plans.

- Can print on large printers.

- High quality printing.

- Plotters are expensive.



Plotter

Speakers

- A speaker, or speaker system converts an electrical signal to sound.

- The term loudspeaker can refer to individual transducers (known as drivers), or to complete systems consisting of a enclosure incorporating one or more drivers and electrical filter components.
- Loudspeakers are the most variable elements in an audio system and are responsible for the greatest degree of audible differences between sound systems.

Speakers



Thank You